

TOWARDS A CHARACTERIZATION OF THE UNCERTAINTY CURVE FOR GRAPHS

Bastien Padeloup Vincent Gripon Grégoire Mercier Dominique Pastor

Télécom Bretagne, UMR CNRS Lab-STICC
Brest, Brittany, France

firstname.lastname@telecom-bretagne.eu

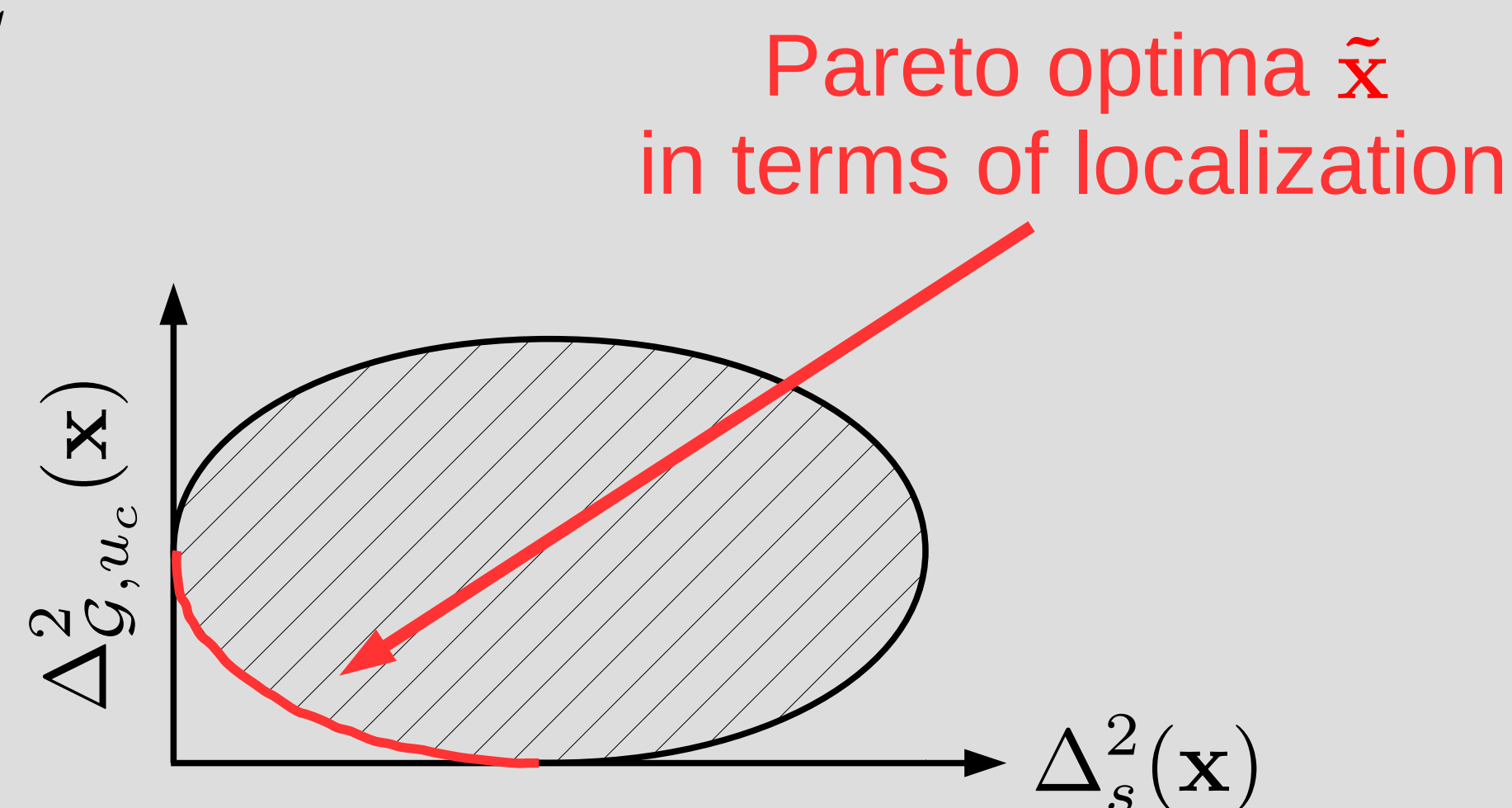
Motivation

Heisenberg's uncertainty principle: $\Delta_t^2(\mathbf{x}) \Delta_\omega^2(\mathbf{x}) \geq C$

A signal $\mathbf{x}$ cannot be fully localized both in time and frequency domains

Agaskar & Lu's uncertainty principle on graphs [1]:

A signal $\mathbf{x}$ cannot be fully localized both in graph and spectral domains



How do we characterize the optima?

Agaskar & Lu's uncertainty principle on graphs

Two essential properties:

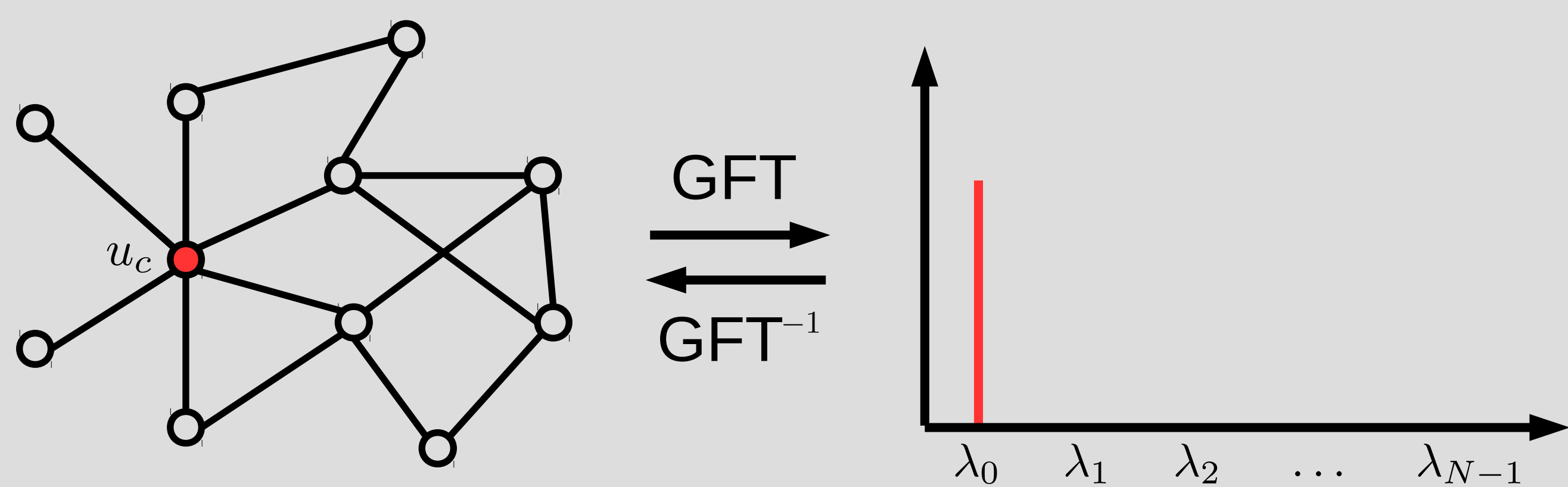
The eigenbasis of the Laplacian matrix $\mathbf{L}$ of a graph carries a notion of frequency and allows the definition of a graph Fourier transform [2]

The multiplicity of eigenvalue $\lambda_0 = 0$ is equal to the number of connected components [3]

Localization measurement tools [1]:

$$\text{Graph spread: } \Delta_{G,u_c}^2(\mathbf{x}) = \sum_{n=0}^{N-1} d(u_c, u_n) \mathbf{x}_n^2$$

$$\text{Spectral spread: } \Delta_s^2(\mathbf{x}) = \sum_{n=0}^{N-1} \lambda_n \hat{\mathbf{x}}_n^2$$



This configuration is impossible, for any graph

Extension to a larger class of graphs

Let $\mathbf{M}(\alpha) = \mathbf{P} - \alpha \mathbf{L}$, for α fixed and $\mathbf{P}$ the diagonal matrix of distances to u_c from every node

If $\mathbf{M}(\alpha)$ is of the following form:

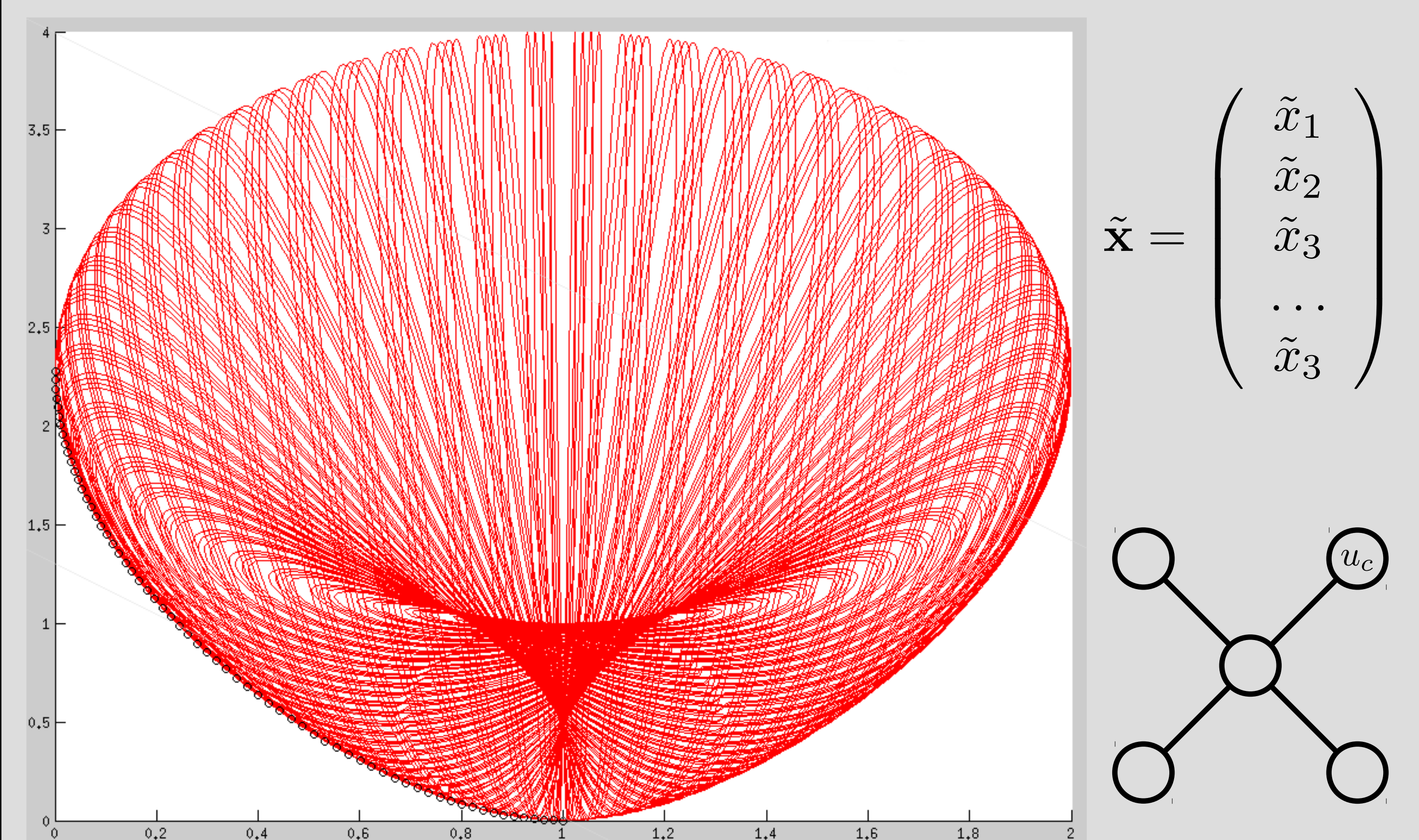
$$\mathbf{M}(\alpha) = \left(\begin{array}{c|c} \mathbf{A} & \mathbf{B} \\ \hline \mathbf{C} & \mathbf{D} \end{array} \right)$$

Constant per line
Circulant, of dimension k

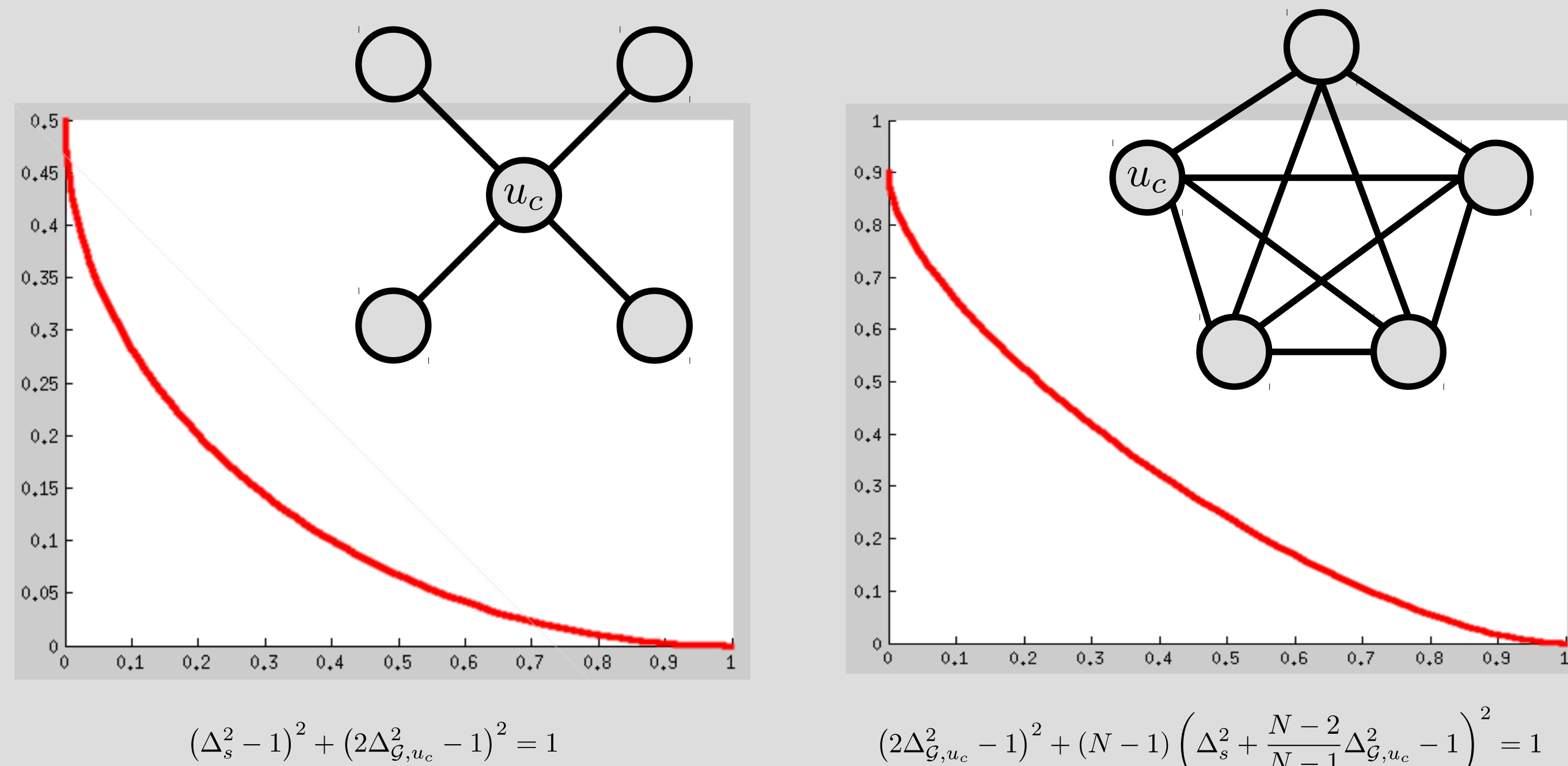
Then $\tilde{\mathbf{x}}$ is the eigenvector associated to the smallest eigenvalue of $\mathbf{M}(\alpha)$, and is of the form

$$\tilde{\mathbf{x}} = \{\tilde{x}_1; \tilde{x}_2; \dots; \underbrace{\tilde{x}_j; \tilde{x}_{j+1}; \dots; \tilde{x}_{j+1}}_{k \text{ times}}\}$$

Illustration on a star graph with excentered u_c



Known uncertainty curves



Optimal signals are of the form $\tilde{\mathbf{x}} = \{\tilde{x}_1; \tilde{x}_2; \dots; \tilde{x}_2\}$ [1]

Conclusions & Future work

Contribution:

Characterization of $\tilde{\mathbf{x}}$ for more complex graphs than [1]

Future work:

Impact of the choice of u_c ?

Given N , is there a non-trivial universal uncertainty curve?

[1] A. Agaskar and Y. M. Lu, "A spectral graph uncertainty principle", CoRR, vol. abs/1206.6356, 2012.

[2] D. I. Shuman, S. K. Narang, P. Frossard, A. Ortega, and P. Vandergheynst, "Signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular data domains", CoRR, vol. abs/1211.0053, 2012.

[3] F. R.K. Chung, "Spectral graph theory", vol. 92, American Mathematical Soc., 1997.